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Course : Artificial Intelligence
Course code : CSA1411

ASSESSMENT TOOL-1-CO-1
SLOT : D
DATE : 10/10/2026

05 10/10/26

1) DEPTH - LIMIT SEARCH

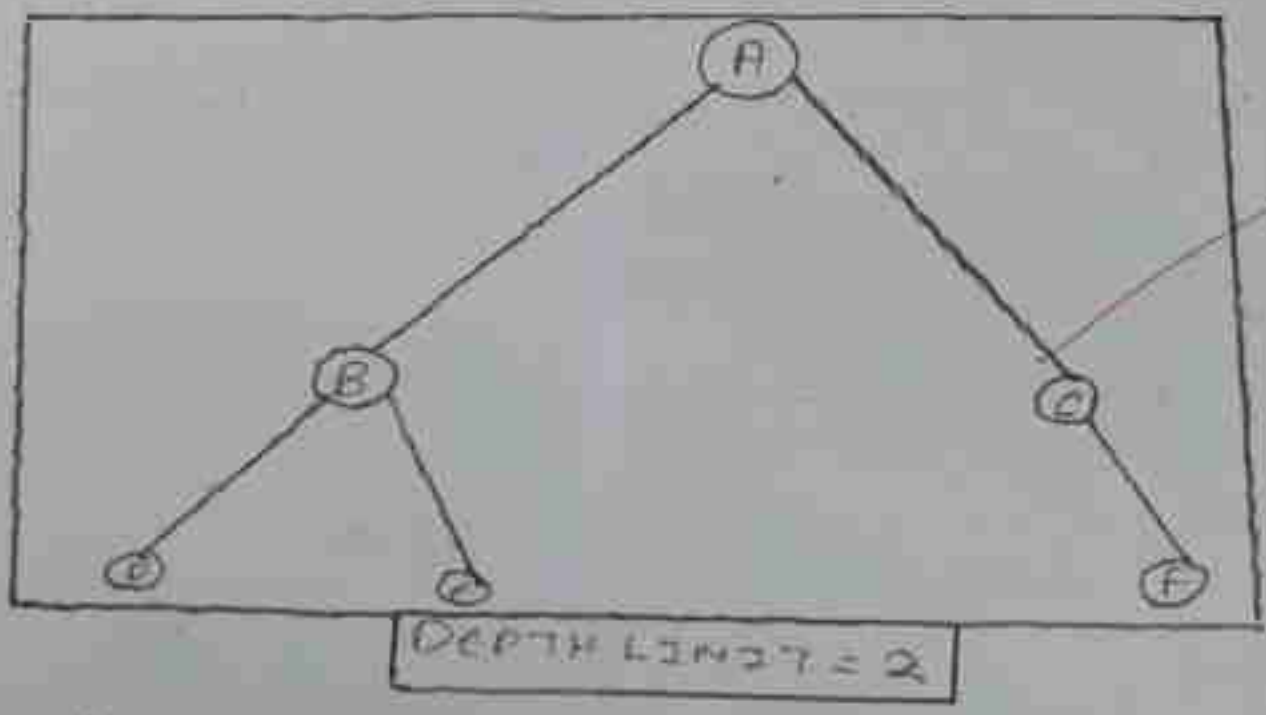
DEFINITION :

DLS is a DFS with depth limit search. It doesn't continue to find the goal when the limit is reached

ALGORITHM / PSEUDOCODE :

1) At first start with an initial goal
2) It continues to check the current search is a goal
3) It search that branch until the limit is reach
4) If it find the goal, it return
5) Otherwise the return is empty set

EXAMPLE :



Traversal : It search until the Depth limit is 2

STATERGY : DATA - LIMIT SEARCH (DLS)

DATA - STRUCTURE : LIFO

RESULT :

COMPLETE : NO, If the limit is reached

OPTIMAL : NO

TIME COMPLEXITY : $O(b^l)$

SPACE COMPLEXITY : $O(bl)$

ADVANTAGE : If works in Infinite branches to find goal

DISADVANTAGE : If the limit is reached it does not gives solution

APPLICATION : It is branch free with depth Search limit

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2) BIDIRECTIONAL SEARCH :

DEFINITION :

Bidirectional Search is a two Search algorithm. First from the forward direction and another from the goal, is backward If two search is meet the goal is achieved

STATERGY : BIDIRECTIONAL SEARCH

DATA - STRUCTURE : Stack FIFO

ALGORITHM / PSEUDOCODE :

1) Start at the first from the forward direction

2) Another from the goal direction is backward

3) Expand the frontiers accordingly

4) If the two search is meet

5) the result is return, otherwise not result ~~dolces~~ place

RESULT :

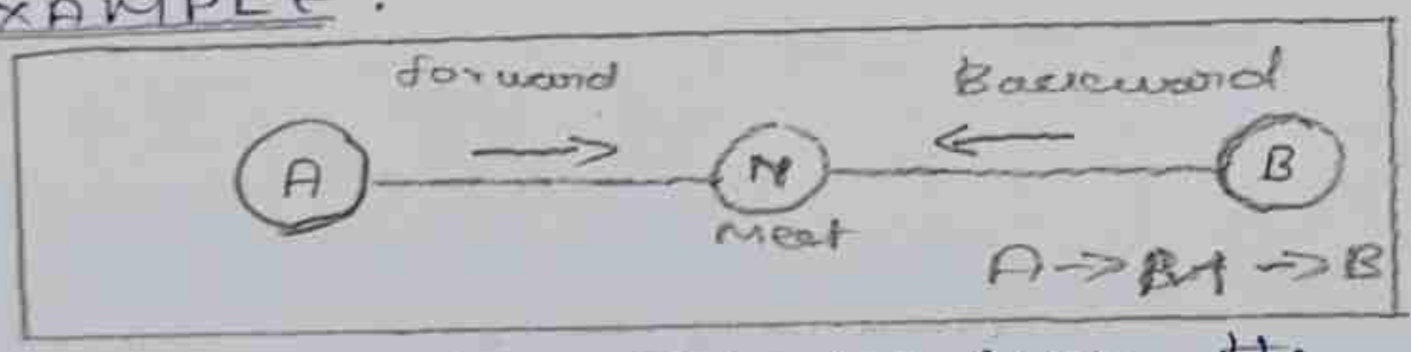
COMPLETE : YES, the result found by the meet of two direction

OPTIMAL : YES

TIME COMPLEXITY : $O(b^{(d/2)})$

SPACE COMPLEXITY : $O(b^{(d/2)})$

EXAMPLE :



ADVANTAGE : It reduce the Search time and find the solution easily

DISADVANTAGE : It required the GOAL and the GOAL from the backward

APPLICATION : It used to find two search algorithm at the time